

FINTECH 540 - Machine Learning for Fintech

Fall Semester 2025

Third Project Checkpoint

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- ▶ We are locating fraudulent transactions. We are able to take unsupervised approaches, but for a supervised approach we would need labeled data from an external source. These sources may have different types of features, different time periods, or missing relevant features compared to our dataset. Is pursuing a supervised approach still realistic and valuable in this context, or would this be a “garbage in, garbage out” situation?
- ▶ Our unsupervised models flag anomalies, but we do not have ground-truth labels. Beyond heuristics or visual inspection, what academically rigorous methods exist to tune or evaluate unsupervised anomaly-detection models in this domain? Are there standard proxy metrics or validation strategies we should consider using?

- ▶ How can we properly create a supervised learning set by classifying data beforehand and thresholding indexes/companies as high or low volatility in an efficient manner?

- ▶ After clustering the assets using RV/RQ-based features, how can we objectively evaluate whether these clusters have real statistical meaning or economic interpretation? Should we add internal metrics (like silhouette score) or external validation (like comparing clusters to market regimes)?

- ▶ Feature engineering with multiple datasets: Our features come from joining zeromev, txs, transfers, account labels, and token labels. Some of these (especially account/token labels) are fairly noisy and may not cover all addresses. Do you have recommendations on how aggressive we should be with filtering—e.g., only using “verified” tokens and clearly labeled DEX addresses—versus keeping more data but accepting label noise?

- ▶ Model evaluation focus: Since our task is highly imbalanced and misclassifying swap MEV may be more costly than misclassifying non-swap MEV, should we focus more on metrics like recall / F1 for the swap class and AUROC, or is overall accuracy still the main evaluation criterion for this project?

- ▶ What should we do to deal with the financial data with different start dates?
- ▶ Should we verify whether missing data points or non-trading days were interpolated, dropped, or forward-filled?

- ▶ 1. The volatility dataset contains multiple realized measures across different frequencies (RV, BPV, Good, Bad, RQ, and their 5-minute versions). We are unsure whether we should: Use all realized measures as features, focus mainly on RV as the target variable, or select only a subset of measures that carry unique predictive information.

- ▶ 2. EDA Findings & Modeling Direction. Our EDA reveals clear volatility clustering and strong correlations among realized measures. We are considering three possible modeling paths: Predicting RV as a regression problem, classifying high-volatility vs low-volatility days, or identifying structural breaks/regime shifts. Which of these directions best aligns with the course expectations?

- Some teammates are exploring external variables (VIX, SPY returns, macro indicators). Our questions are: Is the incorporation of external data encouraged or optional? Are there limits on the amount or type of external data we should use? How should we handle alignment, missing dates, and potential data leakage when merging external data with RV?